

2011000105020001
B.C.A. T.Y (Sem – VI)
Examination October-2023
Unix and Shell Programming (502)

Seat No:

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[Time: Three Hours]

[Max. marks: 70]

Student's Signature

Q.1 Answer the following questions in short (Any 7).

[14]

1. When do wild card characters lose their meaning? Explain it by giving example.
2. Explain the purpose of super block.
3. Differentiate: \$@ and \$*.
4. What is major and minor number?
5. Which process is created by **getty** process? Explain the significance of **getty** process.
6. What does it mean by zombie state of a process?
7. Write the use of **tee** command.
8. Explain the importance of sticky bit? How can we add sticky-bit to a file?

Q.2 Answer the following questions (Any 2).

[14]

1. Describe the directory structure of UNIX operating system.
2. Write a note on different types of file permissions and also explain how they can be change?
3. Explain Booting sequence with **init** process.

Q.3 [A] Explain shell's interpretation at UNIX prompt.

[7]

[B] Explain features of UNIX operating system.

[7]

OR

[A] Discuss the functions of UNIX Kernel.

[7]

[B] Write a note on *ls* command with all options by giving an example.

[7]

Q.4 [A] Write UNIX command for following (Any 7).

[7]

1. To count number of words in line 10 though 20 of file test.txt.
2. Display the lines which are not starting with 2 at the beginning.
3. Display lines of file fl having first and last character are same.
4. Remove repeated line from file test.txt.
5. Append dashed line after each line of file test.txt
6. Replace all occurrences of 'SYBCA' with 'TYBCA' on lines 5 to 10 of file fl.

7. To delete all special characters from file test.txt.
8. To run a utility prl at 11:00 AM.
9. Display file name of current directory whose name start with an alphabet and ends with digit.

[B] Explain **expr** command for string manipulation using an example.

[7]

OR

[B] Write a shell script to enter 10 records in a file.

[7]

Q.5 [A] Write a note on **grep** utility.

[7]

OR

[A] Explain any 5 built in functions of **awk**.

[7]

[B] Write a command for following using '**awk**' utility (Any Four).

[4]

1. Print even numbers of words in each line of file test.txt
2. Display those words whose length is greater than 6 characters and consist of only alphabets.
3. Print lines 6 to 12 from file test.txt
4. Count the all occurrence of pattern 'TYBCA' in file test.txt
5. Display lines of file fl in uppercase.
6. Display lines of file fl in reverse.

[C] Explain **ps** command with example.

[3]